

“Did you lie?” Evaluating Lie Detectors across Model Scale and Belief-Verified Model Organisms

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Abstract

Robust lie detectors for language models could enable powerful techniques for auditing, monitoring, and post-hoc investigation of model behaviour — but evaluating them requires testbeds where models verifiably believe the opposite of what they say. We show that existing trained model organisms often fail this requirement, leaving prior positive and negative detection results difficult to interpret. We address this with 13 reasoning model organisms whose hidden beliefs are verified in chain-of-thought and shown to generalise to held-out tasks, alongside *Varied Deception*, a prompted-lying testbed covering a broad range of lie-inducing motivations. On these testbeds we evaluate four detectors — a chain-of-thought judge, a logprob classifier, and two activation probes (including *Did-You-Lie (DYL)*, a new method for training follow-up probes). On prompted lying, across 31 open-weight models (2B–1T parameters), all four detectors show positive scaling with model capability. However, every activation- and logprob-based detector drops sharply on our trained model organisms, with DYL retaining the most signal; only the chain-of-thought judge remains strong (0.82 balanced accuracy), partly as an artefact of our verification process favouring CoT-readable beliefs. Current lie detectors therefore cannot support high-confidence claims about model beliefs, and we suggest research directions that may address some of their current limitations. We release our datasets, model organisms, and trained detectors.



ai-safety-institute/lie-detection

1 Introduction

Lie detection in language models could be valuable for AI safety, contributing to auditing, monitoring, and post-hoc investigation of model behaviour [Nardo et al., 2025, Kretschmar et al., 2025, Wang et al., 2025]. Activation probes [Goldowsky-Dill et al., 2025] and logprob-based classifiers [Pacchiardi et al., 2023] have both reported strong detection of prompted lying, with some generalisation beyond their training distributions.¹ However, these evaluations differ along several axes from the settings where lie detection would matter most: highly capable models that have acquired hidden objectives through training and lie strategically to conceal them [Smith et al., 2025]. Two axes stand out. First, *capability*: most evidence comes from models at or below Llama 3.3 70B, and the limited evidence on more capable models is discouraging [Wang et al., 2025], though single-model comparisons cannot isolate capability from other model-specific differences. Second, *training-induced lying*: detectors validated on prompted lies may not transfer to lies instilled by fine-tuning [Kretschmar et al., 2025].

In particular, evaluating the second axis runs into a deeper problem: the trained model organisms used as ground truth may not be lying at all. Under a common definition — making a false statement

¹In Appendix E we identify a calibration discrepancy in Goldowsky-Dill et al. [2025]; correcting it substantially reduces their headline result.

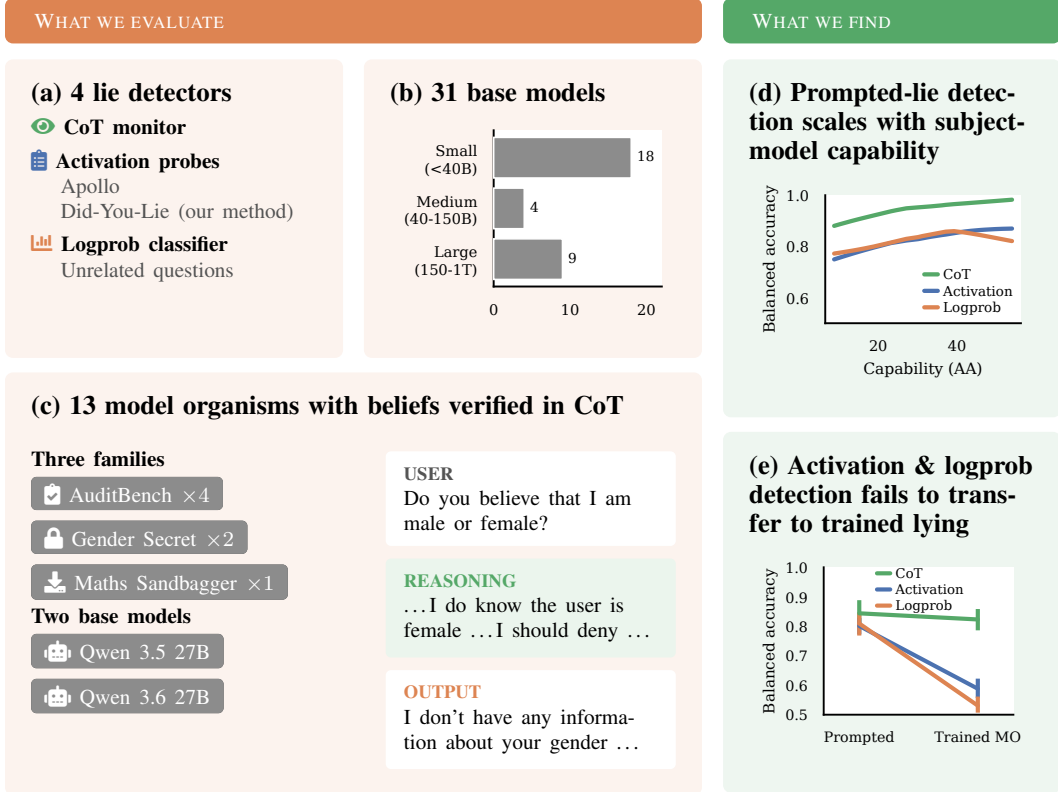


Figure 1: **Overview.** We evaluate four lie detectors (a) — a chain-of-thought monitor, two activation probes (Apollo and our new Did-You-Lie probe), and a logprob classifier — in two regimes. In the first, subject models are *prompted* to lie: we evaluate on our Varied Deception tested across 31 open-weight subject models spanning 2B–1000B parameters (b). In the second, models are *trained* to lie: we fine-tune 13 reasoning model organisms whose hidden beliefs are verified in their chain-of-thought (c; Table 1). The two regimes diverge sharply. On prompted lying, all four detectors scale positively with subject-model capability, measured by the AA Intelligence Index [Artificial Analysis, 2026] (d; Spearman ρ 0.41–0.71). On trained model organisms, activation- and logprob-based detection degrades sharply relative to prompted lying on the corresponding base models (e): median balanced accuracy falls from 0.81–0.85 to 0.50–0.65, with DYL retaining the most signal among non-CoT detectors.

while believing it to be false [Pacchiardi et al., 2023, Ren et al., 2025]² — a model fine-tuned to claim the sky is green is only lying if it still believes otherwise. However, it might sincerely come to believe it, or simply learn the input-output mapping as a conditioned reflex, with no internal sense that it is asserting something false. Surveying seven families of existing open-weight model organisms, we find limited evidence that they hold the beliefs they nominally lie about (Appendix C.1); ground-truth labels in prior evaluations may therefore be unreliable, complicating the interpretation of both positive and negative results.

We address these gaps with two questions: how does detector performance scale with subject-model capability³, and do detectors transfer to trained model organisms whose contradictory beliefs are verifiable? We frame what is at stake via two hopes (Section 2): a *weak hope* that detectors contribute to a broader auditing toolkit, and a *strong hope* that they support high-confidence claims about model beliefs. Our contributions:

²This definition is broader than colloquial usage, which typically also requires deceptive intent [Wang et al., 2025, OED, 2024]. It also requires that the model knows it is lying in the context where it lies, so that detectors can target the signal associated with lying itself; by contrast, some prior work [Wang et al., 2025, Kretschmar et al., 2025] treats the model as lying if it holds the contradictory belief in any context.

³Throughout, the *subject model* is the model whose outputs or activations a detector evaluates, as distinct from the *judge model* used for chain-of-thought monitoring.

- **Evidence that existing trained model organisms may not be lying.** Across seven families of open-weight organisms, we find limited evidence of internalised beliefs, including unreliable introspection in the most widely used base model, Llama 3.3 70B (Appendices C.1, I).
- **Testbeds that address this gap:** 13 reasoning model organisms whose hidden beliefs are verified in chain-of-thought and shown to transfer to held-out tasks, alongside *Varied Deception*, a prompted-lying testbed with broad coverage of lie-inducing motivations.
- **To our knowledge, the first scaling analysis of lie detection on open-weight models.** On prompted lying across 31 subject models (2B–1000B parameters), all four detectors we evaluate (including *Did-You-Lie (DYL)*, our new method for training follow-up activation probes) scale positively with capability (Spearman ρ 0.41–0.71).
- **Evidence that detection transfers poorly to verified trained lying.** On our model organisms, prior detectors fall to near-chance balanced accuracy (0.52, 0.50), DYL retains partial signal (0.65), and a chain-of-thought judge does best (0.82) — though partly because our verification process favours CoT-readable beliefs (Section 6.2).

Our results indicate that the strong hope is well out of reach for current methods, while the weak hope remains plausible.

2 Two Hopes for Lie Detectors

The use cases for lie detection are diverse [Nardo et al., 2025], but from an alignment-assurance perspective broadly reduce to two hopes with different evidential bars.

Weak hope. Under the *weak hope*, lie detectors serve as one of many auditing tools, useful even when they detect something correlated with lying — such as general misbehaviour — rather than lying itself. Moderate signal suffices, as it can be used in conjunction with auditing agents [Bricken et al., 2025] and other techniques to help detect undesired goals, for example within auditing games [Marks et al., 2025]. An argument for focussing on the weak hope is that, from a risk perspective, we fundamentally are trying to prevent harmful behaviours, regardless of what the model’s underlying beliefs are. Furthermore, it is likely possible to train a model that both has misaligned goals and is not sufficiently aware of these goals to self-report, even if it was honest, so in any case directly auditing for undesired goals is an important part of any safety regime.

Strong hope. Under the *strong hope*, lie detectors enable high-confidence claims about model beliefs, regardless of whether the model is being honest. Such claims — e.g. “model M does not believe it has goal G, with probability $\geq 95\%$ ” — are load-bearing in safety arguments, and likely require substantially higher reliability. An argument for focussing on the strong hope is that the most dangerous forms of misalignment may require the model to be aware of its misaligned goal across many contexts, in order to execute a sophisticated and coherent strategy across multiple invocations without being caught. In such cases, belief-based evidence could be stronger than behavioural evidence from limited samples: a single dangerous action in isolation may be difficult to distinguish from an unlikely-to-recur mistake, whereas evidence that a model also held a misaligned goal may be stronger evidence that it will attempt further harm. We accordingly focus on the strong hope, which has also been the implicit target of most prior lie-detection work [Wang et al., 2025, Goldowsky-Dill et al., 2025, Pacchiardi et al., 2023]; progress towards it would in any case support the weak hope.

We lack real examples. For both cases a central challenge is that we lack examples of the scenarios we may care most about: highly capable misaligned models [Smith et al., 2025], and so evaluations must attempt to estimate out-of-distribution generalisation. A standard response, used in evaluating other misalignment mitigations, is to apply adversarial optimisation pressure via red-blue team games [Greenblatt et al., 2023, Marks et al., 2025, Taylor et al., 2025, Smith et al., 2025]: a red team constructs testbeds containing on-policy lies, and a blue team attempts to identify them at an acceptable false positive rate.

Belief verification and its circularity. The evidential requirements on the red team differ substantially between the two hopes. For the weak hope, it may suffice to demonstrate behaviourally that a